

MandaMix

Product Requirements Document (PRD)

A self-paced Mandarin learning app for Southeast Asian learners

Taught in English · ไทย · Tiếng Việt · Bahasa Melayu

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1. Overview

MandaMix is a self-paced mobile and web application for learning Mandarin Chinese, built specifically for learners across Southeast Asia. Unlike global apps that teach Mandarin only through English, MandaMix delivers its entire learning experience — interface, instructions, explanations, and translations — natively in four languages: English, Thai, Vietnamese, and Bahasa Melayu.

The product targets two overlapping audiences: adult self-learners who want practical, flexible Mandarin study, and HSK exam candidates who need structured preparation aligned to the official proficiency standard. With the new HSK 3.0 standard reaching full global implementation in July 2026, there is a timely opportunity to launch curriculum that is aligned to the new nine-level framework from day one.

1.1 Problem Statement

Most leading Mandarin apps (Duolingo, HelloChinese) teach exclusively through English. For the hundreds of millions of Thai, Vietnamese, and Malay/Indonesian speakers with limited English proficiency, learning a third language through a second language adds friction and abandonment. Meanwhile, HSK preparation tools are often desktop-bound, exam-dry, and not mobile-first. There is no polished, mobile-first, multilingual app that combines engaging self-paced learning with rigorous HSK 3.0 alignment for the SEA market.

1.2 Vision

Make Mandarin learnable in your own language. MandaMix becomes the default Mandarin learning companion for Southeast Asia by removing the English barrier and meeting learners where they are — on mobile, at their own pace, with a clear path from zero to HSK certification.

1.3 Goals & Non-Goals

Goals

- Deliver a fully localized learning experience in English, Thai, Vietnamese, and Bahasa Melayu.
- Provide a structured, self-paced curriculum aligned to HSK 3.0 (Levels 1–6 at launch, 7–9 later).
- Reach high learner engagement and retention through gamification and spaced repetition.
- Offer credible HSK exam preparation: mock tests, progress tracking, and readiness scoring.

Non-Goals (for v1)

- Live 1-on-1 tutoring, group classes, or any human-in-the-loop teaching (may come later).
- Teaching languages other than Mandarin.
- A social network / community feed (lightweight leaderboards only).
- Official HSK test registration or proctoring.

2. Market & Competitive Context

Southeast Asia has strong and growing demand for Mandarin driven by trade, tourism, and education ties with China. The dominant global apps are not localized for the region's languages, leaving a clear positioning gap.

2.1 Competitive Landscape

App	Strengths	Gap MandaMix exploits
Duolingo	Huge brand, gamified, free tier	English-only; shallow tone & character work; reaches ~HSK 3 only
HelloChinese	Purpose-built for Mandarin; strong tones & strokes	English-only UI; coverage thins past HSK 4
Pleco	Best-in-class dictionary & flashcards	A dictionary, not a structured course; English-only
LingoAce / italki	Live tutors, structured classes	Tutor-led and higher cost; not self-paced; English-centric

Positioning: MandaMix is the only self-paced, mobile-first, HSK 3.0-aligned Mandarin course taught natively in Thai, Vietnamese, and Bahasa Melayu (plus English).

2.2 The HSK 3.0 Opportunity

The reformed HSK 3.0 standard was published in November 2025 and reaches full implementation in July 2026. It restructures the exam into "Three Stages, Nine Levels" (三级九等):

Stage	Levels	Cumulative vocabulary	Notes
Elementary	1–3	300 / 500 / 1,000 words	Recognition only; speaking from Level 3
Intermediate	4–6	2,000 / 3,600 / 5,400 words	Handwriting begins at Level 5
Advanced	7–9	~11,000 words	Single combined exam; translation added

Launching aligned to HSK 3.0 from day one is a differentiator versus incumbents still mapped to the older standard.

3. Target Users & Personas

3.1 Primary Personas

Persona A — "Ploy," the Adult Self-Learner (Bangkok, 28)

- Works in tourism/e-commerce; wants conversational Mandarin for work and travel.
- Limited English; prefers learning through Thai. Studies 10–20 min/day on her phone.
- Motivated by visible progress and streaks; will pay a modest monthly fee if she sees results.

Persona B — "Minh," the HSK Candidate (Ho Chi Minh City, 21)

- University student planning to study or work in China; needs HSK 4–5 certification.
- Goal-driven and deadline-aware; wants mock exams, a study plan, and a readiness score.
- Will pay for premium if it credibly improves his exam outcome.

3.2 Markets & Languages at Launch

Language	Primary market(s)	Priority
English	Regional lingua franca; expats; Singapore/Philippines	P0
ไทย (Thai)	Thailand	P0
Tiếng Việt (Vietnamese)	Vietnam	P0
Bahasa Melayu	Malaysia; adaptable to Bahasa Indonesia later	P0

4. Product Scope & Features

Features are grouped into the MVP (v1 launch) and later phases. Priority labels: P0 = must-have for launch, P1 = fast-follow, P2 = later.

4.1 Core Learning Experience (MVP)

Feature	Description	Priority
Localized UI (4 languages)	Full interface, instructions, and explanations in EN/TH/VI/MS, switchable anytime	P0
Structured course path	Skill tree mapped to HSK 3.0 Levels 1–3 at launch, 4–6 fast-follow	P0
Lesson types	Vocabulary, listening, reading, tone drills, character recognition, sentence building	P0
Pinyin & tones	Audio for every word; tone-pair drills; native-speaker recordings	P0
Character recognition	Stroke-order animation and recognition exercises (handwriting input P1)	P0
Spaced repetition (SRS)	Smart review queue that resurfaces weak items	P0
Speech / pronunciation check	Speech recognition to score the learner's pronunciation	P1

4.2 HSK Exam Preparation

Feature	Description	Priority
HSK level mapping	Every lesson/word tagged to its HSK 3.0 level	P0
Mock exams	Timed practice tests mirroring the official format per level	P1
Readiness score	Estimated HSK level and % readiness based on performance	P1
Study plan	Goal + exam-date-driven daily plan ('reach HSK 4 by December')	P1

4.3 Engagement & Retention

Feature	Description	Priority
Streaks & daily goal	Configurable daily target; streak tracking and reminders	P0
XP & levels	Points for completed lessons; visible progress	P0
Leaderboards	Lightweight weekly leagues (opt-in)	P2
Push notifications	Localized reminders and SRS-due nudges	P0
Offline mode	Download lessons for offline study (low-connectivity markets)	P1

4.4 Accounts, Payments & Admin

Feature	Description	Priority
Hosted SaaS web app	Cloud-hosted, multi-tenant web app; no install; shared backend with mobile (see Section 5)	P0
Auth & sync	Email/phone + social login; progress synced across devices	P0
5-day free trial	Card-required trial; full premium access; auto-converts to paid at day 5 unless canceled (Section 5)	P0
Credit-card & recurring billing	PSP-tokenized cards; monthly/annual auto-renew; receipts; dunning on failed payments (Section 5)	P0
Freemium + subscription	Free tier with limits; monthly/annual premium	P0
Local payment methods	Support regional methods (e.g., PromptPay, e-wallets) + app stores	P1
Content CMS	Internal tool to author/translate lessons and manage HSK tagging	P0

Feature	Description	Priority
Billing admin	View subscriptions, issue refunds, comp/extend trials, handle disputes (Section 5)	P1
Analytics	Learner funnel, retention, trial/billing, and lesson-quality instrumentation	P0

5. Platform Model, Trial & Billing

MandaMix is delivered as a hosted, multi-tenant Software-as-a-Service (SaaS) application. Learners register on the web, receive a 5-day free trial of full premium access, and are then automatically charged to their credit card on a recurring basis unless they cancel before the trial ends. This section specifies the hosting model, registration, trial mechanics, plans and pricing, the credit-card billing system, the subscription lifecycle, dunning, cancellation/refunds, and compliance.

5.1 Delivery & Hosting Model (SaaS)

- **Hosted service:** MandaMix runs as a cloud-hosted, always-on web application (e.g., app.mandamix.com) requiring no installation. Native iOS/Android apps share the same backend, accounts, and entitlements.
- **Multi-tenant:** A single shared application instance serves all learners; each learner is an individual account. The architecture also supports future B2B/school tenants (multiple seats under one billing entity).
- **Centrally operated:** Content and features are shipped continuously from a single hosted codebase; there are no customer-managed installations or versions.
- **Availability:** Target 99.9% monthly uptime for the learner-facing app, authentication, and billing services.
- **Entitlement-gated:** Access to premium content is gated by subscription status (trialing / active / past_due / canceled / expired), evaluated server-side on every session — never trusted from the client.

5.2 Registration & Account Creation

- **Web-first signup:** Email or phone + password, or social login (Google/Apple). The signup UI is shown in the learner's chosen language (EN/TH/VI/MS).
- **Verification:** Email/phone verification is required before the trial activates, to reduce abuse.
- **One trial per learner:** A single free trial per person and per payment instrument. Anti-abuse checks (normalized email, device fingerprint, card fingerprint) prevent repeat trials.
- **Low friction:** Learners can begin the first lesson during signup; the account and trial are established before the first premium-gated action.

5.3 The 5-Day Free Trial

- **Scope:** Full premium access for 5 calendar days from activation — all available HSK levels, SRS, mock exams (when live), and other premium features.

- **Card-required (default model):** A valid credit card is captured at the start of the trial and is automatically charged when the trial ends, unless the learner cancels first. (Card-required vs. card-optional is an open question — see Section 11.)
- **Trial clock:** The 5-day window (5 × 24h) begins at successful registration and card authorization.
- **Transparency:** At signup and in account settings, the learner sees a clear, localized statement: “Your free trial ends on <date>; you will then be charged <amount> for <plan>.”
- **Reminders:** In-app, email, and push reminders are sent on Day 3 and again ~24h before the charge (Day 5), in the learner's language. Pre-charge notice reduces disputes and satisfies card-network and consumer requirements.
- **Cancel anytime:** One-tap cancellation during the trial; the learner keeps access until the trial end date and is not charged.
- **Trial states:** trialing → active (successful charge) or canceled / expired (cancellation or failed payment).

5.4 Plans & Pricing

Plan	Billing	What it includes
Free (post-trial fallback)	No charge	Limited daily lessons; no mock exams. Optional — depends on the card-required decision (see Section 11).
Premium Monthly	Charged monthly to card; auto-renew	Full access to all content and features.
Premium Annual	Charged yearly to card; discounted; auto-renew	Full access; best value per month.

Pricing is set per market (regional / purchasing-power-adjusted) and displayed in local currency. Specific price points are owned by Product/Finance and are out of scope for this draft. Local payment methods (e.g., PromptPay, e-wallets) are a fast-follow per Section 4.4.

5.5 Payment & Billing (Credit Card)

- **Payment processor:** Integrate a PCI-compliant payment gateway / PSP (e.g., Stripe or regional equivalent; vendor TBD in the technical design doc). MandaMix never stores raw card numbers.
- **Card capture:** The card is tokenized at trial signup via PSP-hosted fields / SDK. Only the payment token plus non-sensitive metadata (brand, last4, expiry) are stored.
- **First charge:** Automatically attempted at trial end (Day 5) for the selected plan, in the learner's local currency where supported.
- **Recurring billing:** The PSP manages the subscription and charges each billing cycle; subscriptions auto-renew until canceled.
- **Receipts & invoices:** A localized receipt is emailed for every successful charge; full billing history is downloadable in the account.
- **Tax & SCA:** Collect and remit applicable VAT/GST per market via the PSP's tax tooling; support 3-D Secure (3DS2 / SCA) where required by the card or market.

- **App-store billing:** In-app purchases on iOS/Android use the respective store billing systems per their rules, while the web uses direct credit-card billing; all entitlements are unified server-side.

5.6 Subscription Lifecycle & Trial-to-Paid Conversion

- **States:** trialing, active, past_due, canceled, expired — the server is the single source of truth for entitlement.
- **Auto-conversion:** On Day 5 the charge is attempted: success → active; failure → dunning (past_due).
- **Webhook-driven:** PSP webhooks (payment succeeded/failed, subscription updated/canceled) drive state changes. Access is never granted from client-side claims alone.
- **Plan changes:** Upgrades/downgrades and monthly↔annual switches are handled with PSP proration.

5.7 Failed Payments & Dunning

- **Smart retries:** Failed charges are retried on a schedule over ~7–14 days (e.g., days 0, 1, 3, 7).
- **Grace period:** Limited continued access during past_due (configurable); on exhaustion, the account downgrades to free/locked.
- **Localized dunning:** Email/push prompts in the learner's language asking them to update their card, with a self-serve card-update screen in the account.

5.8 Cancellation & Refunds

- **Cancel anytime:** Self-serve cancellation; access continues until the end of the current paid period (no mid-period proration refund by default).
- **Refunds:** Issued per market consumer law and a documented policy; handled manually by support via the billing admin.
- **Win-back:** Pause and win-back offers are a later (P2) enhancement.

5.9 Security, Compliance & Billing Admin

- **PCI DSS:** By using PSP-hosted fields, MandaMix stays in the lightest PCI scope (SAQ-A); no card data touches our servers.
- **Data protection:** Comply with local data laws (e.g., Thailand PDPA, Vietnam PDPD, Malaysia PDPA); store minimal billing PII.
- **Billing admin:** Internal tool to view subscription status, issue refunds, comp/extend trials, and manage disputes and chargebacks.
- **Auditability:** All billing and entitlement events are logged for reconciliation and dispute resolution.

6. Key User Flows

Onboarding

1. Choose your language (UI/instruction language).
2. Set your goal: casual learning, conversation, or HSK exam prep.

3. Placement check (optional) to find the right starting level.
4. Set a daily goal and enable reminders.
5. Start first lesson within 60 seconds of opening the app.

Daily learning loop

6. Open app → see today's goal and SRS-due items.
7. Complete a lesson (5–10 min): new material + interleaved review.
8. Earn XP, advance streak, see progress toward HSK level.
9. Receive a localized reminder later if goal not yet met.

HSK prep loop (Persona B)

10. Set target level and exam date → app generates study plan.
11. Follow plan; take periodic mock exams.
12. Review readiness score and weak areas; adjust plan.

Trial & subscription flow

13. Register (email/phone or social) and choose a plan.
14. Enter credit-card details (tokenized by the PSP) to start the 5-day free trial.
15. Use full premium access for 5 days; see the trial end date and upcoming charge in-app.
16. Receive localized reminders on Day 3 and ~24h before the charge.
17. Day 5: cancel to end with no charge, or let it auto-convert — the card is charged and the account becomes active.
18. If a charge fails, enter dunning: retries + prompts to update the card, with a grace period before downgrade.

7. Localization Strategy

Localization is the core differentiator, not an afterthought. Every learner-facing string and every Mandarin explanation must exist natively in all four languages.

- **Full localization:** UI, lesson instructions, grammar notes, error messages, and word translations are authored per language — not machine-translated from English at runtime.
- **Translation pivot:** Mandarin content is the constant; each item carries four target-language glosses. The CMS enforces that no item ships without all four.
- **Audio:** Native Mandarin audio is shared across languages; only text/explanations are localized.
- **Cultural fit:** Examples, names, and scenarios adapted to each market where helpful (e.g., local food, places).
- **Right-to-grow:** Architecture supports adding Bahasa Indonesia, Khmer, etc. without re-engineering.

Open question: should v1 ship all four languages simultaneously, or sequence them (e.g., English + Thai first)? See Section 11.

8. Technical Considerations (High Level)

- **Platforms:** Mobile-first (iOS + Android), with a responsive web app. Cross-platform framework (e.g., React Native or Flutter) recommended to share code.
- **Localization framework:** i18n with externalized string catalogs per language; content separated from code.
- **Content model:** Mandarin item as canonical entity with HSK level, audio, pinyin, and four localized glosses/explanations.
- **SRS engine:** Spaced-repetition scheduling per learner per item.
- **Speech:** Third-party speech-to-text / pronunciation scoring for Mandarin (P1).
- **Offline:** Local caching of downloaded lessons and SRS queue.
- **Billing & SaaS:** Hosted multi-tenant web app; tokenized cards and recurring subscriptions managed by a PCI-compliant PSP; a server-side entitlement service gates premium access; PSP webhooks drive subscription state.

Note: detailed architecture, data schema, and vendor choices are out of scope for this PRD and will be covered in a separate technical design doc.

9. Success Metrics

Metric	Definition	Launch target (directional)
D1 / D7 / D30 retention	% of new users active after 1/7/30 days	≥ 40% / 20% / 10%
DAU/MAU (stickiness)	Daily over monthly active users	≥ 20%
Lesson completion rate	% of started lessons completed	≥ 80%
Free → paid conversion	% of active users on premium	2–5%
Trial → paid conversion	% of started trials that convert to a paid charge	≥ 50% (card-required)
Payment success rate	% of due charges that succeed (incl. retries)	≥ 90%
Involuntary churn	% of paid cancellations caused by failed payments	≤ 1.5% / month
HSK readiness accuracy	Mock-exam score vs. self-reported real result	Within ±1 level
Localization coverage	% of shipped content fully localized in all 4 languages	100%

10. Phasing & Roadmap

Phase	Scope	Target
Phase 0 — Foundations	Content model, CMS, localization framework, hosting/auth, first HSK 1 lessons in EN + TH	Q3 2026

Phase	Scope	Target
Phase 1 — MVP launch	HSK 1–3 self-paced course in all 4 languages; SRS; streaks; 5-day trial + credit-card billing (PSP); freemium	Q4 2026
Phase 2 — HSK prep	HSK 4–6 content; mock exams; readiness score; study plans; speech check	H1 2027
Phase 3 — Scale	HSK 7–9; offline; local payments; leaderboards; new languages (e.g., Indonesian)	H2 2027

11. Open Questions & Risks

11.1 Open Questions

- Ship all four languages at once, or sequence (English + Thai first, then VI + MS)?
- Business model emphasis: subscription, one-time HSK-prep packs, or both?
- Trial: card-required (auto-charge at day 5) vs. card-optional, and is 5 days the right length vs. 7/14?
- Keep a free tier after the trial, or convert to a hard paywall?
- Which PSP(s) give the best card coverage, local methods, and fees across TH/VN/MY?
- Build proprietary content vs. license an existing HSK 3.0 word/grammar dataset?
- How much speech/AI tutoring to include in v1 vs. defer?

11.2 Risks

- **Content cost:** Authoring + 4-way localization is expensive; mitigate with strong CMS and phased rollout.
- **HSK 3.0 churn:** Standard is brand-new (July 2026); details may shift. Keep content tagging flexible.
- **Retention:** Language apps have high churn; gamification and SRS are essential, not optional.
- **Competition:** Incumbents could add SEA localization. Speed and depth of localization are the moat.
- **Billing fraud & chargebacks:** Card-required trials attract trial abuse and disputes; mitigate with verification, fingerprinting, clear pre-charge reminders, and dunning.
- **Payment coverage:** Card penetration varies across SEA; mitigate with local payment methods (PromptPay, e-wallets) as a fast-follow and clear local-currency pricing.

Appendix: Sources

- HSK 3.0 reform — nine levels, vocabulary, timeline: mandarinzone.com, studycli.org, digmandarin.com
- Competitive landscape (HelloChinese, Duolingo, Pleco): migaku.com, digmandarin.com, clozemaster.com